

ATMA: Long-Context Language Modeling via Polar Attention and Gated-Delta Compression Memory

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Abstract

Zero-shot length extrapolation requires a sequence model to balance exact retrieval, long-document likelihood, and inference cost. We introduce ATMA, a hybrid architecture whose Polar Attention separates a radially normalized direction channel from a bounded participation-ratio magnitude channel and augments both with gated-delta recurrent memory. We evaluate long-context modeling as a Pareto problem rather than claiming universal architectural dominance. A 120-run, 1B-token factorial sweep selects the recipe; matched 10B-token, 378M-parameter NoPE, RoPE, and Polar models, plus stateful Raven baselines, are then evaluated through 256K tokens. Polar preserves real-text retrieval at 64K (16.3% versus 0.0% for NoPE and RoPE) and limits 256K bits-per-byte inflation to $1.26\times$, with a 1.5-point mean penalty on short-context tasks. Raven instead leads 256K BABILong and offers constant-state decoding. Finally, an independent environment audit finds that NoPE runs with nearly identical 2K validation curves differ by 6.70 nats at 256K across GPU device classes. This exposes unresolved interactions among execution environment, optimization, and extrapolation, motivating multi-objective and multi-environment audits of long-context claims.

1 Introduction

Language models increasingly process repositories, books, and scientific documents far longer than their training sequences. Standard softmax attention retains explicit access to old tokens, but its dense normalization can disperse mass as the key population grows (Nakanishi, 2025; Velickovic et al., 2025; Barbero et al., 2024). A complementary extreme-value view observes that the largest chance match among unrelated keys grows with context length (Huang et al., 2026). Sliding windows bound the active population but discard distant targets; recurrent mixers keep fixed state but compress the past. Long-context modeling therefore presents several competing objectives: retrieval fidelity, document-level likelihood, reasoning, short-context quality, and serving cost.

We introduce ATMA, a 378M-parameter hybrid decoder that combines local gated convolutions with four global Polar Attention layers. Polar adds a learned null competitor that grows on the $\sqrt{\log n}$ extreme-value scale, radially normalizes the matched value sum, and exposes effective match count through a bounded participation-ratio channel. A gated-delta memory supplies a third, recurrent channel. The design separates “what matched,” “how much matched,” and compressed history while keeping every channel bounded under explicit conditions (Appendix A).

Our study makes three contributions. First, a complete 120-cell, 1B-token factorial sweep isolates the interaction between attention core, recurrent memory, local windows, distractor alignment, and regularization. Second, matched 10B-token attention runs and strong Raven baselines are compared on retrieval, BABILong, fixed-target likelihood, eight short-context tasks, and serving. Figure 1 summarizes the resulting Pareto frontier: Polar leads the attention variants in long-context retrieval and stability, Raven leads 256K BABILong and decode efficiency, and softmax retains a small short-context advantage. Third, an independent controlled diagnostic finds that NoPE checkpoints with effectively identical 2K

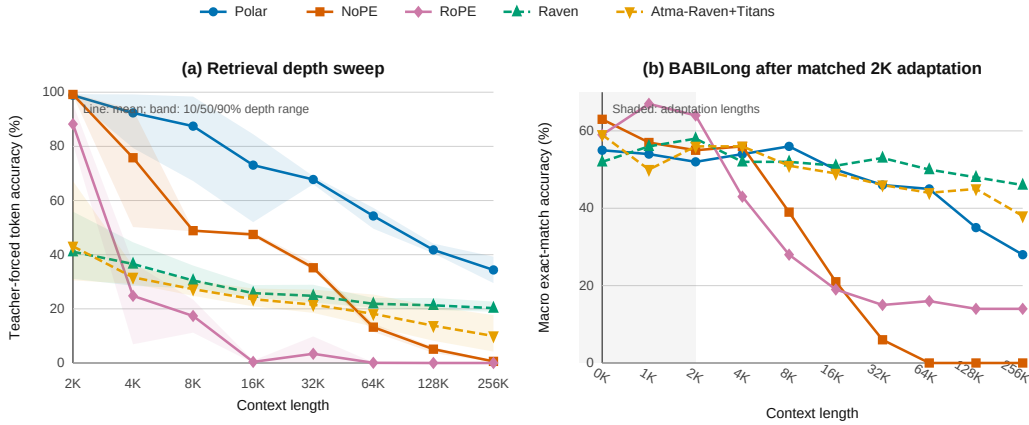


Figure 1: Matched 10B-token results. Left: teacher-forced retrieval accuracy, averaged over retrieval tasks, haystack suites, and needle depths; bands span depth-wise means. Right: BABILong macro exact match after identical answer-only adaptation. Polar leads matched attention variants at long lengths; Raven leads long-context reasoning but changes both architecture family and optimizer.

training curves can differ by 6.70 nats at 256K after training on different GPU classes. We report this as evidence of environment sensitivity, not as proof that reduction order is the unique cause, and call for multi-device replication before architectural attribution.

2 Related Work

Softmax attention (Vaswani et al., 2017) has quadratic cost and may lose sharpness as the number of competitors increases. RoPE (Su et al., 2021) and ALiBi (Press et al., 2021) alter positional treatment but not dense simplex normalization. Scalable-Softmax, adaptive-temperature analyses, and sparse alternatives explicitly study dispersion or size-dependent sharpness (Nakanishi, 2025; Velickovic et al., 2025; Vasylenko et al., 2026). Threshold Differential Attention (TDA) instead uses a $\sqrt{\log n}$ threshold motivated by maxima of noise scores (Huang et al., 2026). Polar adopts this scale but uses a differentiable null sink rather than hard survivor selection.

Linear attention, state-space models, and gated recurrences exchange explicit token access for compressed state (Gu & Dao, 2023; Yang et al., 2024b;a; Behrouz et al., 2025). Raven combines routed sparse memory, decay dynamics, and gated-delta recurrence (Afzal et al., 2026); we treat it as a strong sub-quadratic baseline, but not as an attention-only ablation because its optimizer and model family differ. Hybrid models combine local and global mixers for efficiency (Hasani et al., 2025; Allen-Zhu, 2025). ATMA follows this hybrid pattern while changing the global readout and adding a recurrent memory channel.

3 Method

3.1 Hybrid architecture

ATMA is a 16-layer decoder with 12 LFM2-style gated convolutional layers and four global attention layers. All blocks use pre-normalization and squared-ReLU MLPs. The matched NoPE, RoPE, and Polar variants share hidden size 1024, eight query heads, two key-value heads, head dimension 128, tokenizer, data order, optimizer, schedule, and training budget; only the attention core changes. Canon depthwise causal convolutions are applied to q , k , and v before scoring (Allen-Zhu, 2025).

3.2 Polar attention

For query i and its $n_i = i + 1$ visible keys, let $\sigma_{ij} = q_i^\top k_j / \sqrt{d_k}$. Each head learns a length temperature and a virtual null logit,

$$t_i = 1 + \text{softplus}(\alpha) \log n_i, \quad \nu_i = b + \text{softplus}(\gamma) \sqrt{\log(n_i + 1)}. \quad (1)$$

The second term tracks the extreme-value scale of unrelated scores. Appending the null as a softmax competitor gives

$$w_i = \text{softmax}(t_i[\sigma_{i\bullet}, \nu_i]), \quad s_i = \sum_j w_{ij} v_j + w_{iN} v_{\text{null}}. \quad (2)$$

The direction channel is $c_i = s_i / \max(\|s_i\|_2, \varepsilon)$, hence $\|c_i\|_2 \leq 1$. To retain bounded information about match multiplicity, real-key weights are renormalized as $\hat{w}_{ij} = w_{ij} / \sum_k w_{ik}$ and summarized by the participation ratio

$$n_{\text{eff},i} = \left(\sum_j \hat{w}_{ij}^2 \right)^{-1}, \quad m_{\text{eff},i} = n_{\text{eff},i}(1 - w_{iN}), \quad \mu_i = \tanh(\text{softplus}(\beta) \log(1 + m_{\text{eff},i})). \quad (3)$$

Thus $\mu_i \in [0, 1)$, while the direction channel removes radial scale. This is a channel decomposition, not an independence claim: changing the match set can rotate c_i . The Polar output is a gated projection of c_i plus a learned projection of μ_i . A distractor-alignment auxiliary loss was tested but is disabled in the selected recipe because the factorial sweep shows worse retrieval.

3.3 Gated-delta memory

Each global layer also maintains a per-head associative state $M_t \in \mathbb{R}^{d_v \times d_k}$. With unit-normalized keys and queries, retention γ_t , and write gate β_t , the decay-first, self-inclusive recurrence is

$$M_t = \gamma_t M_{t-1} (I - \beta_t k_t k_t^\top) + \beta_t v_t k_t^\top, \quad qquad r_t = M_t q_t. \quad (4)$$

RMS-normalized r_t is gated, projected, and added to the Polar output. The projection is zero-initialized. Unit L_2 normalization is important: RMS-normalized keys have $\|k\|_2^2 = d_k$ and made the tested delta recurrence diverge, whereas unit keys keep the update eigenvalue along k within the gate-controlled range. Under bounded values and a uniformly contractive retention gate, Appendix A gives a sequence-length-independent state bound. Our empirical claim is narrower: the implemented state remains stable in the tested length sweep.

Figure 2 summarizes the two mechanisms and their controlled diagnostics. Appendix B defines every regularization mode in the factorial sweep, Appendix C gives the numerical oracle and streaming Triton implementation, and Appendix H details the kernel and serving optimizations.

4 Experimental Design

Recipe selection. Stage I trains all $3 \times 5 \times 2 \times 2 \times 2 = 120$ combinations of attention type (Polar, RoPE, NoPE), regularization, distractor loss, memory, and a 1024-token window. Models receive approximately 1B FineWeb-Edu tokens at length 2048 on NVIDIA L4 GPUs. This stage selects a recipe; it is not used for headline absolute comparisons.

Matched comparison. Stage II trains 378.16–378.22M-parameter NoPE, RoPE, and Polar models for 9.816B tokens at length 2048 on the same L40S software and hardware environment. All use the same Muon/AdamW split. Raven Native (388.54M) and Atma-Raven-Titans (382.37M) use the same data, tokenizer, budget, and device class, but Raven’s native AdamW optimizer. This choice follows a 1B-token Raven Native pilot with the same Atma Muon schedule: validation loss improved to 3.76 nats at step 625, then destabilized and ended at 4.62, with 0% retrieval through 64K. We therefore retain Raven’s native AdamW recipe and report both Raven models as a separate baseline group; the single failed pilot does not establish that Raven is incompatible with every Muon configuration.

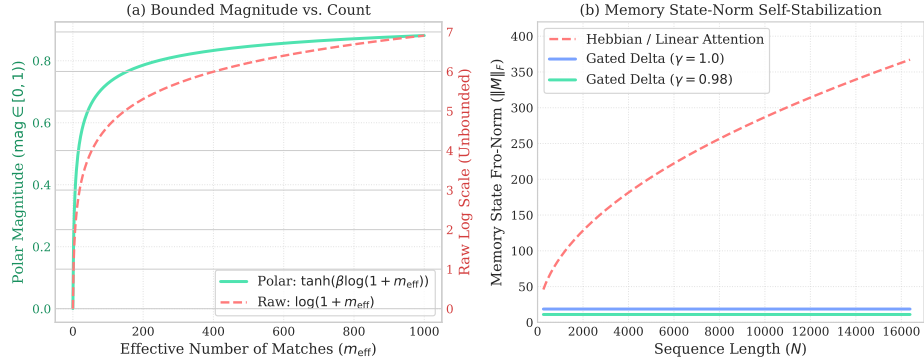


Figure 2: ATMA mixer diagnostics. Left: the effective match count is mapped to a bounded magnitude channel. Right: the tested gated-delta state remains flat with length, unlike an ungated Hebbian memory. These controlled diagnostics illustrate the mechanism; they are not a network-level stability proof.

1B recipe	NLL 2K	NLL 64K	Needle 64K
NoPE + memory	2.665	2.342	16.3%
RoPE + memory	2.808	2.857	0.0%
Polar, no memory	2.826	3.604	0.0%
Polar + memory	2.703	1.960	92.5%
Polar + memory + distractor	2.725	1.975	58.8%
Polar + memory + window	2.712	2.006	30.0%

Table 1: Illustrative 1B-token recipe selection. NLL is nats/token; Needle is greedy per-digit accuracy over 16 five-digit trials.

Evaluation. The suite contains 24,000 paired retrieval trials over synthetic and FinePDFs haystacks, lengths 2K–256K, depths 10/50/90%, and 50 trials per cell; BABILong answer-only adaptation and held-out exact match (Kuratov et al., 2024); fixed-target bits per byte (BPB) on FinePDFs, PG-19, and Proof-Pile; eight likelihood-scored short-context tasks (22,939 examples per model); and single-sequence prefill/decode on one L40S. Retrieval is teacher-forced five-token accuracy, which is less brittle than exact-value match but should not be read as free-running generation success. BABILong uses a custom split and prompts, so its absolute scores are internally comparable rather than leaderboard replications. Protocol details and complete matrices appear in Appendix D.

5 Results

5.1 Component selection

Table 1 shows the decisive transitions. Neither memoryless Polar nor memory-augmented softmax retrieves reliably at 64K. Combining Polar with gated-delta memory raises per-digit retrieval from 0.0% to 92.5%; a local window or distractor alignment then reduces it. We therefore promote global Polar plus memory without the auxiliary distractor loss. The complete 120-cell matrix is in Appendix E.

5.2 A multi-objective frontier

Table 2 reports matched endpoints. At 2K, NoPE and Polar both retrieve almost perfectly. By 256K, NoPE and RoPE fall to 0.6% and 0.0%, while Polar retains 34.4% averaged across tasks, suites, and depths. The aggregate hides an important boundary: on natural FinePDFs haystacks, Polar retains 16.3% at 64K versus 0.0% for NoPE and RoPE, but falls to 0.3% at 256K; on synthetic filler it retains 68.4%. We therefore claim a substantially

Group	Model	Retrieval (%)		BABILong (%)		BPB		
		2K	256K	2K	256K	2K	256K	ratio
Matched	NoPE	99.2	0.6	55	0	1.432	8.097	5.65×
Matched	Polar	98.9	34.4	52	28	1.451	1.825	1.26×
Matched	RoPE	88.2	0.0	64	14	1.439	2.512	1.75×
Raven/AdamW	Atma-Raven-Titans	43.1	10.0	56	38	1.525	1.551	1.02×
Raven/AdamW	Raven Native	41.1	20.3	58	46	1.581	1.597	1.01×

Table 2: Long-context endpoints. Retrieval averages tasks, suites, and depths. BABILong is macro exact match. BPB averages three fixed-target datasets; lower is better. Bold is best within each comparison group.

Model	Haystack	2K	4K	8K	16K	32K	64K	128K	256K
Polar	Synthetic	99.6	98.4	98.5	97.0	94.9	92.3	80.3	68.4
	FinePDFs	98.6	86.3	77.9	49.1	40.7	16.3	3.2	0.3
NoPE	Synthetic	99.5	98.7	98.9	94.9	70.3	26.5	10.2	1.2
	FinePDFs	99.3	52.9	0.0	0.0	0.0	0.0	0.0	0.0
RoPE	Synthetic	81.7	44.9	33.1	0.8	6.8	0.1	0.0	0.0
	FinePDFs	94.8	4.7	0.2	0.0	0.0	0.0	0.0	0.0
Raven Native	Synthetic	55.9	57.7	51.4	48.3	44.3	40.9	38.9	35.1
	FinePDFs	22.1	15.4	8.2	3.3	5.3	2.9	3.7	5.4

Table 3: Zero-shot retrieval accuracy (%) by haystack type. Values average teacher-forced token accuracy over needle depths. The recurrent hybrid is omitted for space and reported in Appendix F.

extended effective window, not solved arbitrary-length retrieval. Full depth and haystack matrices are in Appendix F.

Table 3 separates the retrieval results by haystack and length. Polar also limits mean BPB inflation to 1.26× at 256K, compared with 5.65× for NoPE and 1.75× for RoPE. This stability accompanies a short-context cost: mean accuracy over eight 2K controls is 43.29% for Polar, 45.15% for NoPE, and 44.60% for RoPE. Figure 3 shows that the gap is task-dependent rather than uniform; the 1.86-point mean gap to NoPE prevents a blanket quality claim. Exact per-task values are in Appendix G.

Table 4 makes the short-context quality and systems trade-off explicit. Raven reaches 46% BABILong at 256K and keeps BPB almost flat, while its fixed recurrent state avoids length-dependent KV-cache reads. At 128K, Atma-Raven-Titans decodes at 2.27 ms/token versus 16.3 ms/token for Polar, but Polar has higher synthetic retrieval and higher training MFU. Because the stable reported Raven runs require their native AdamW recipe rather than the failed matched-schedule pilot, these results locate an operational frontier rather than isolate one mechanism.

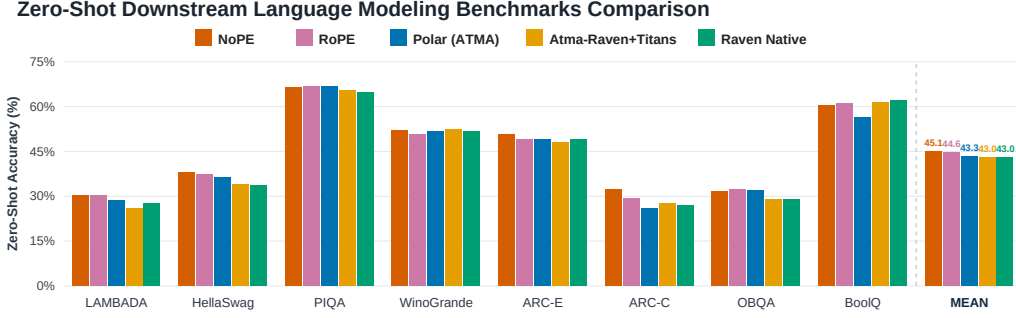


Figure 3: Zero-shot accuracy on eight 2K language-modeling controls (22,939 examples per model). The rightmost bars are unweighted task means. Exact values and metric definitions are in Table 9.

Group	Model	Base mean (%)	128K prefill (s)	Decode (ms/token)	Peak alloc. (GiB)	Train MFU (% , 2K)
Matched	NoPE	45.1	1.47	16.4	39.40	42.77
Matched	Polar	43.3	1.62	16.3	39.41	40.44
Matched	RoPE	44.6	1.47	16.5	39.41	42.45
Raven/AdamW	Atma-Raven-Titans	43.1	1.00	2.27	8.62	30.92
Raven/AdamW	Raven Native	43.0	1.36	3.27	6.46	21.53

Table 4: Short-context and systems trade-offs. Serving uses one sequence and one timing sample on one L40S, so the values are descriptive rather than uncertainty estimates.

6 Environment Sensitivity and Open Questions

Beyond the architecture comparison, Figure 4 presents a separate empirical contribution: nearly identical short-context convergence can conceal sharply different long-context behavior. Three NoPE runs finish at 2.8126, 2.8128, and 2.8160 validation nats, yet their 256K NLLs are 10.77 (L40S, microbatch 16), 9.90 (L40S, microbatch 4), and 4.07 (L4, microbatch 4).

Diagnostic journey. The finding emerged through an infrastructure transition rather than a planned device comparison. Stage I and the first Stage II 10B NoPE control ran on L4 with microbatch 4 and produced the 4.07-nat endpoint. Moving the promoted runs to L40S with microbatch 16 preserved the 2K trajectory but changed the 256K endpoint to 10.77. Re-evaluation under PyTorch 2.12 and 2.13 left both checkpoints’ scores unchanged; an L40S microbatch-4 retrain still reached 9.90; and deterministic 2K stream fixtures showed no gradient or loss discrepancy. These steps exclude evaluation-version and microbatch-only explanations in the tested runs, but do not isolate a unique hardware or kernel cause. All five headline Stage II models were trained together on L40S, so the main comparison remains environment-matched; Appendix I records the full audit sequence.

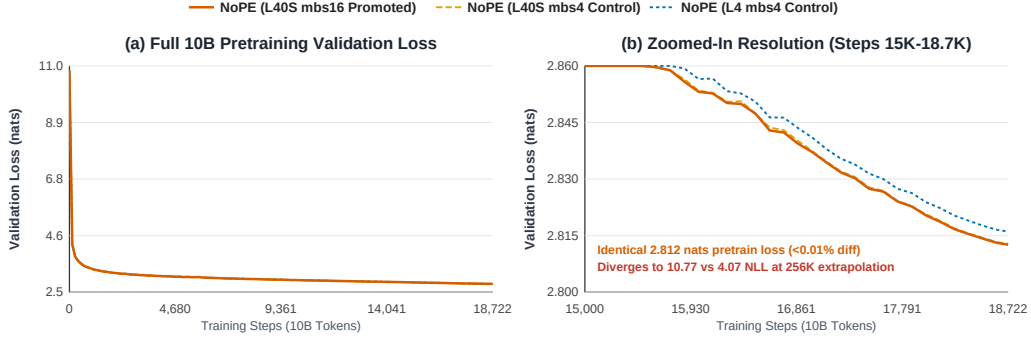


Figure 4: NoPE 2K validation trajectories are nearly coincident, yet the checkpoints’ 256K NLL values are 10.77, 9.90, and 4.07. The two L40S runs differ in microbatch size; the L4 run differs in device class.

FEA Structural Stress Probing: Local Block Secant Gains G Across Length (2K -> 256K)

Polar (ATMA): Flat Gains Across Length									RoPE: Representational Attenuation									NoPE (L40S mbs16): Block 7 Yield Locus								
	2K	4K	8K	16K	32K	64K	128K	256K		2K	4K	8K	16K	32K	64K	128K	256K		2K	4K	8K	16K	32K	64K	128K	256K
B0	0.86	0.86	0.86	0.86	0.86	0.86	0.86	0.86	B0	0.94	0.94	0.94	0.94	0.94	0.94	0.94	0.94	B0	0.91	0.91	0.91	0.91	0.91	0.91	0.91	0.91
B1	0.78	0.78	0.78	0.78	0.78	0.78	0.78	0.78	B1	0.92	0.92	0.92	0.92	0.92	0.92	0.92	0.92	B1	0.88	0.88	0.88	0.88	0.88	0.88	0.88	0.88
B2	0.96	0.95	0.95	0.95	0.96	0.96	0.96	0.96	B2	1.01	1.00	0.99	0.98	0.98	0.98	0.98	0.98	B2	1.02	1.02	1.02	1.02	1.02	1.03	1.03	1.03
B3	0.90	0.89	0.89	0.89	0.88	0.88	0.88	0.87	B3	1.02	1.02	1.02	1.02	1.02	1.02	1.02	1.02	B3	0.98	0.98	0.98	0.97	0.97	0.97	0.97	0.97
B4	0.87	0.87	0.86	0.86	0.86	0.85	0.85	0.85	B4	1.05	1.07	1.08	1.07	1.06	1.04	1.03	1.03	B4	0.96	0.96	0.96	0.96	0.96	0.96	0.96	0.95
B5	0.86	0.85	0.85	0.84	0.84	0.84	0.84	0.84	B5	0.94	0.93	0.93	0.93	0.93	0.93	0.93	0.92	B5	0.94	0.94	0.94	0.94	0.95	0.96	0.96	0.96
B6	1.02	1.02	1.03	1.05	1.07	1.09	1.10	1.11	B6	1.09	1.06	1.04	1.02	1.01	1.01	1.00	1.00	B6	1.13	1.11	1.09	1.09	1.08	1.09	1.09	1.10
B7	0.98	0.97	0.97	0.97	0.97	0.97	0.97	0.97	B7	1.01	1.01	1.00	1.00	1.00	1.00	0.99	0.99	B7	1.02	1.02	1.02	1.03	1.09	1.20	1.31	1.38
B8	0.96	0.96	0.95	0.95	0.95	0.95	0.95	0.94	B8	1.01	1.00	1.00	1.00	0.99	0.99	0.99	0.99	B8	1.01	1.01	1.01	1.00	1.00	1.00	1.00	0.99
B9	0.93	0.92	0.92	0.92	0.92	0.92	0.92	0.91	B9	0.98	0.98	0.98	0.97	0.97	0.97	0.97	0.96	B9	0.99	0.99	0.99	0.98	0.98	0.98	0.98	0.99
B10	1.10	1.10	1.10	1.10	1.10	1.10	1.10	1.09	B10	1.09	1.08	1.07	1.06	1.06	1.05	1.04	1.04	B10	1.16	1.14	1.12	1.12	1.12	1.11	1.11	1.11
B11	1.02	1.02	1.02	1.02	1.02	1.01	1.01	1.01	B11	1.02	1.02	1.02	1.01	1.01	1.01	1.01	1.00	B11	1.04	1.03	1.03	1.02	1.01	1.01	1.01	1.00
B12	1.02	1.01	1.01	1.01	1.01	1.00	1.00	1.00	B12	1.02	1.02	1.01	1.01	1.01	1.01	1.00	1.00	B12	1.04	1.04	1.03	1.02	1.01	1.01	1.01	1.00
B13	1.01	1.01	1.00	1.00	1.00	1.00	1.00	1.00	B13	1.02	1.01	1.01	1.00	1.00	1.00	1.00	1.00	B13	1.04	1.04	1.06	1.08	1.09	1.07	1.06	1.06
B14	1.43	1.44	1.45	1.44	1.41	1.39	1.38	1.37	B14	1.32	1.31	1.29	1.28	1.27	1.26	1.25	1.25	B14	1.39	1.37	1.35	1.35	1.30	1.26	1.24	1.23
B15	1.23	1.23	1.22	1.21	1.20	1.19	1.18	1.17	B15	1.21	1.20	1.19	1.19	1.18	1.17	1.17	1.16	B15	1.18	1.18	1.17	1.14	1.12	1.11	1.09	1.08

Figure 5: Sampled block secant gains across context length. The probe localizes the L40S NoPE checkpoint’s change near blocks 6–7, while Polar remains comparatively flat. A sampled gain is a directional diagnostic, not a tight global Lipschitz estimate.

Figure 5 localizes the observed checkpoint change with a sampled block-level stress probe. The evidence establishes sensitivity to the training environment, but it does not uniquely identify CUDA reduction order as the causal mechanism: device class is not randomized or replicated across multiple seeds, and other hardware-dependent kernels may contribute. The defensible conclusion is that short-context convergence did not predict long-context behavior in these runs. Architecture-level extrapolation claims should therefore report seeds, kernels, device classes, and extended-context audits; stronger causal claims require replicated cross-device interventions. Additional diagnostic controls are in Appendix I.

What this audit cannot answer. The finding opens several concrete questions that the present compute budget and page limit do not resolve:

1. Does the device-class divergence replicate across seeds, software stacks, and models at 7B scale and above, or is it specific to this 378M setup?
2. Which training-level numerical difference—kernel selection, reduction or accumulation order, or optimizer-state evolution—is amplified into the divergent endpoint, and can an early diagnostic predict it?

3. Does Polar’s 0.3% natural-text retrieval at 256K improve with longer or mixed-domain training, or reflect a synthetic-to-natural induction bias?
4. How does the learned null competitor interact with gated-delta memory under heterogeneous pretraining data?
5. How much of Raven’s 256K BABILong lead comes from recurrent state, sparse routing, or AdamW, given that the matched Muon schedule did not converge?

These questions delimit the current evidence rather than extend its claims; resolving them requires interventions that separately vary architecture, optimizer, data, and execution environment.

7 Limitations and Conclusion

The study uses one model scale, one 2K pretraining regime, and fewer independent training replicates than needed for population-level uncertainty. Teacher-forced retrieval, custom BABILong adaptation, and single-sample serving measurements answer different questions and should not be collapsed into one score. Polar still fails on natural-text retrieval at 256K, its theoretical bounds require learned-margin assumptions that are not yet empirically certified for every head, and Raven is not an optimizer-matched ablation.

Within those limits, ATMA demonstrates that bounded full-context attention plus recurrent memory can move the long-context Pareto frontier. Polar materially extends soft retrieval and stabilizes long-document likelihood while retaining competitive short-context quality; Raven illustrates the reasoning and serving advantages of compressed state. The environment diagnostic further shows why long-context claims need evaluation well beyond the training window and, where feasible, across hardware environments. We release configurations, logs, checkpoint manifests, benchmark matrices, and proof artifacts to make each reported comparison auditable.

AI Use Statement

Generative AI tools were used in this revision to edit and shorten the manuscript, condense the presentation of existing mathematical claims and proofs, improve readability, and suggest document structure. They were not used to generate experimental datasets or execute experiments in this revision. The authors will manually compare all AI-assisted mathematical text with the original derivations and proof artifacts, verify retained empirical claims against source logs and cited work, and take responsibility for every claim and artifact in the submission. This statement describes the present revision only; before submission, the authors will disclose any additional AI-assisted research tasks from the complete project history as required by the ICLR 2027 policy.

Reproducibility Statement

Model definitions and training settings are described in Sections 3–4. Appendix D specifies evaluation protocols and denominators; Appendices E–I provide detailed tables and diagnostics. The anonymous artifact at <https://anonymous.4open.science/r/atma> contains configurations, logs, checkpoint manifests, benchmark payloads, and machine-checked proof skeletons.

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A Theoretical Framing

These are sufficient-condition results. They establish noise-odds and norm bounds under explicit assumptions; they do not establish distributional invariance or replace the empirical evidence.

A.1 Extreme-value null floor

Assumption 1 (Sub-Gaussian noise scores). For irrelevant keys $j \in \mathcal{N}_n$, $|\mathcal{N}_n| \leq n$, scores X_j are mean-zero and v^2 -sub-Gaussian: $\mathbb{E} \exp(tX_j) \leq \exp(v^2 t^2/2)$ for all $t \in \mathbb{R}$.

Let the Polar null floor be $\nu_n = b + s\sqrt{\log(n+1)}$ with $b, s \geq 0$.

Proposition 1 (Noise exceedances). Under Assumption 1,

$$\mathbb{E} \left[\sum_{j \in \mathcal{N}_n} \mathbf{1}\{X_j > \nu_n\} \right] \leq n \exp\left(-\frac{\nu_n^2}{2v^2}\right). \quad (5)$$

If $s \geq \sqrt{2}v$ this is $O(1)$; if the inequality is strict it decays polynomially.

Proof. Apply the sub-Gaussian tail bound to each key and sum. Since $b \geq 0$, $\nu_n^2 \geq s^2 \log(n+1)$. $\square$

Unlike TDA, Polar retains sub-threshold keys. With temperature $\theta_n = 1 + a \log n$, define the unnormalized noise-to-null odds $R_n = \sum_{j \in \mathcal{N}_n} \exp(\theta_n(X_j - \nu_n))$.

Proposition 2 (Vanishing null odds). Suppose $s > c > \sqrt{2}v$ and the null floor is $s\sqrt{\log(n+1)}$ up to a nonnegative offset. With probability tending to one,

$$R_n \leq n \exp\left(-(1 + a \log n)(s - c)\sqrt{\log n}\right). \quad (6)$$

For $a > 0$, the bound converges to zero faster than any inverse polynomial.

Proof. A union bound gives $\Pr[\max_j X_j > c\sqrt{\log(n+1)}] \leq n(n+1)^{-c^2/(2v^2)} \rightarrow 0$. On the complementary event, every noise score is at least $(s-c)\sqrt{\log n}$ below the floor. Bound each summand and sum over at most n keys. $\square$

A.2 Bounded channels and recurrent state

Proposition 3 (Bounded Polar readout). For $c_i = s_i / \max(\|s_i\|_2, \varepsilon)$, $\|c_i\|_2 \leq 1$. For $\mu_i = \tanh(\beta \log(1 + m_{\text{eff},i}))$, $\mu_i \in [0, 1)$ when $\beta, m_{\text{eff},i} \geq 0$. A distribution uniform over m real keys has participation ratio m .

Proof. The first two claims follow from the denominator and range of $\tanh$. Uniform weights satisfy $(\sum_{j=1}^m (1/m)^2)^{-1} = m$. $\square$

Assumption 2 (Contractive memory). For every t , $\|q_t\|_2 = \|k_t\|_2 = 1$, $\|v_t\|_2 \leq V$, $0 \leq \beta_t \leq 1$, and $0 \leq \gamma_t \leq \bar{\gamma} < 1$.

Proposition 4 (Length-bounded state). For recurrence equation 4,

$$\|M_t\|_F \leq \bar{\gamma}^t \|M_0\|_F + \frac{V}{1 - \bar{\gamma}}. \quad (7)$$

Proof. $I - \beta_t k_t k_t^\top$ has spectral norm at most one and $\|\beta_t v_t k_t^\top\|_F \leq V$. Thus $\|M_t\|_F \leq \bar{\gamma} \|M_{t-1}\|_F + V$; unrolling gives the result. $\square$

The Lean artifact checks the finite-sum, exponential-monotonicity, bounded-channel composition, and invariant-set induction steps. It takes the sub-Gaussian tail, high-probability margin, and one-step contraction as premises. It is therefore a machine-checked proof skeleton, not end-to-end formal verification. The learned conditions $b \geq 0$ and $s > c > \sqrt{2}v$ require an empirical head-wise audit before they can be asserted for a trained checkpoint.

B Regularization Modes

The factorial grid varies a representation regularizer applied to the model’s signature stream. For every non-baseline mode, the sweep uses the same weight $\alpha_{\text{sig}} = 0.01$ and optimizes

$$\mathcal{L} = (1 - \alpha_{\text{sig}})\mathcal{L}_{\text{LM}} + \alpha_{\text{sig}}\mathcal{L}_{\text{reg}} + \lambda_{\text{dist}}\mathcal{L}_{\text{dist}}. \quad (8)$$

The weak and strong modes follow the two SIGReg families introduced by Weak-SIGReg (Akbar, 2026) and LeJEPA’s strong SIGReg objective (Balestriero & LeCun, 2025). The discrete and zipfian modes are local variants that combine those ideas with stronger geometric priors; Table 5 defines all five sweep settings.

Mode	Meaning in the sweep
baseline	No signature-stream regularization; equivalently, $\alpha_{\text{sig}} = 0$.
weak	Weak SIGReg: center a random sketch of the token representations and match its covariance to the identity, $\ \text{Cov}(Sx) - I\ _F$. This encourages decorrelated, unit-variance features at low cost.
strong	Strong SIGReg: project representations onto random one-dimensional directions and match their empirical characteristic functions to the standard Gaussian characteristic function over a fixed frequency grid.
discrete	Local variant: normalize each representation to the sphere of radius $\sqrt{d}$, then apply weak covariance matching. This preserves whitening pressure while encouraging fixed-norm, discrete-code-like directions.
zipfian	Local variant: normalize directions for an orthogonality penalty, then match sorted representation magnitudes to a Zipf profile r^{-1} . This biases the signature stream toward heavy-tailed feature use rather than uniform energy allocation.

Table 5: Regularization modes in the 120-cell factorial sweep. Weak and strong are SIGReg-style covariance and distribution-matching objectives; discrete and zipfian are exploratory local variants.

C Polar Attention Implementation

Numerical oracle. The materialized PyTorch path forms the masked real-key logits and virtual null logit in FP32, applies a single softmax, and computes the direction and magnitude channels from the resulting weights. It is used as a numerical oracle for forward and backward tests, not as the production long-context path, because materializing the $T \times T$ score matrix is memory-prohibitive at the evaluated lengths.

Streaming sufficient statistics. The Triton kernel follows the online-max strategy used by FlashAttention-style kernels. For one query row, it streams K, V blocks and retains only four FP32 statistics: the running maximum M , real-key exponential sum L , squared exponential sum Q_2 , and unnormalized value sum S . If a new block changes the row maximum, $\alpha = \exp(M_{\text{old}} - M_{\text{new}})$ rescales each statistic as shown in Table 6.

State	Online update	Use after streaming
M	$M \leftarrow \max(M, \max a)$	Stable common logit origin
L	$L \leftarrow \alpha L + \sum_j p_j$	Real-key softmax mass
Q_2	$Q_2 \leftarrow \alpha^2 Q_2 + \sum_j p_j^2$	$n_{\text{eff}} = L^2 / Q_2$
S	$S \leftarrow \alpha S + \sum_j p_j v_j$	Direction numerator

Table 6: Streaming state for one Polar query row, where a is the current logit block and $p_j = \exp(a_j - M_{\text{new}})$. The squared accumulator must rescale by α^2 .

After the real keys are consumed, the virtual null is folded in as one additional softmax entry. Let $p_0 = \exp(t_i v_i - M)$ and $Z = L + p_0$. The kernel obtains

$$c_i = \text{normalize}(S + p_0 v_{\text{null}}), \quad n_{\text{eff},i} = \frac{L^2}{Q_2}, \quad m_{\text{eff},i} = n_{\text{eff},i} \frac{L}{Z}. \quad (9)$$

The common factor $1/Z$ cancels under radial normalization of the direction numerator. The production kernel evaluates the magnitude with $2\sigma(2x) - 1 = \tanh x$ and saves (M, L, Q_2, S) for the backward pass. Listing 1 gives the algorithmic core; pointer arithmetic, launch metadata, and backward code are omitted.

Listing 1: Schematic core of the block-streamed Triton forward pass.

```
@triton.jit
def polar_fwd_core(q, K, V, n_i, temp, nullv, beta,
                  BLOCK_N: tl.constexpr, DK: tl.constexpr):
    M = tl.full([BLOCK_M], -1e38, tl.float32)
```

```

L = tl.zeros([BLOCK_M], tl.float32)
Q2 = tl.zeros([BLOCK_M], tl.float32)
S = tl.zeros([BLOCK_M, DK], tl.float32)

for start in range(0, n_i, BLOCK_N):
    k, v, valid = load_kv_block(K, V, start, n_i)
    a = tl.where(valid, temp[:, None] * dot(q, k), -1e38)
    M_new = tl.maximum(M, tl.max(a, axis=1))
    alpha = tl.exp(M - M_new)
    p = tl.where(valid, tl.exp(a - M_new[:, None]), 0.0)
    L = alpha * L + tl.sum(p, axis=1)
    Q2 = alpha * alpha * Q2 + tl.sum(p * p, axis=1)
    S = alpha[:, None] * S + tl.dot(p, v)
    M = M_new

# Fold in the learned virtual null as one extra entry.
M_new = tl.maximum(M, temp * nullv)
alpha = tl.exp(M - M_new)
L, Q2, S = alpha * L, alpha * alpha * Q2, alpha[:, None] * S
p_null = tl.exp(temp * nullv - M_new)
Z = L + p_null

direction = normalize(S + p_null[:, None] * v_null)
n_eff = L * L / tl.maximum(Q2, eps)
m_eff = n_eff * L / tl.maximum(Z, eps)
magnitude = 2 * tl.sigmoid(2 * beta * tl.log(1 + m_eff)) - 1
save_for_backward(M_new, L, Q2, S)

```

This removes the quadratic score-matrix storage; full-context prefill still performs quadratic query-key arithmetic. The kernel is cross-checked against the materialized PyTorch oracle in FP32 accumulation before lower-precision outputs are accepted.

D Evaluation Protocols

Stage I contains 120 approximately 370–378M-parameter runs trained for 1,900 steps of 524,288 tokens at sequence length 2,048 on FineWeb-Edu. Stage II contains three matched attention models trained for 18,722 such steps (9.816B tokens). The five promoted checkpoints and ten open-weight controls are evaluated with checkpoint-exact forward paths.

Raven optimizer control. The Raven rows use Raven’s native AdamW recipe because the attempted matched-schedule control did not produce a usable baseline. In the 1B-token Raven Native pilot with the Atma Muon schedule, validation loss decreased from 10.83 to 3.76 nats by step 625, then rose to 6.88 at step 1,125 and ended at 4.62; teacher-forced retrieval was 0% at every tested length from 2K through 64K. The reported Raven Native and Atma-Raven-Titans checkpoints therefore use AdamW and remain a separate model-family/optimizer group. This is evidence about the tested Muon schedule, not a general claim that Raven cannot converge under any Muon tuning.

Retrieval crosses passkey and NIAH tasks, synthetic and FinePDFs haystacks, eight lengths from 2K through 256K, three depths, and 50 paired trials per cell. The primary metric is teacher-forced accuracy over five target tokens. BABILong uses answer-only fine-tuning on tasks qa1–qa10 at 0K/1K/2K and greedy exact match on ten held-out rows per task and length. Base-task controls use LAMBADA, HellaSwag, PIQA, WinoGrande, ARC-Easy, ARC-Challenge, OpenBookQA, and BoolQ. Fixed-target BPB uses FinePDFs, PG-19, and Proof-Pile. All reported processes completed without failed, OOM, unsupported, missing, or null cells.

E Complete Factorial Sweep

Table 7 retains all 120 factorial cells; the main paper reports only the recipe transitions needed to justify promotion.

Table 7: Complete Results of the 120-Run Ablation Sweep. All models are 370M parameters, trained on FineWeb-Edu for 1B tokens ($seq_len = 2048$). Evaluated from 2K to 64K context length. Clean NLL is measured on coherent FinePDFs documents; junk NLL is measured on the concatenated validation stream. NLL is in nats/token (lower is better), Needle is greedy per-digit accuracy in % (higher is better), and MFU is training model flops utilization in %.

Type	Reg	Dist	Mem	Win	Val	Clean2K	Clean64K	Junk2K	Junk64K	Ndl2K	Ndl64K	MFU
POLAR	baseline	Off	Off	Off	3.323	2.83	3.60	3.27	5.68	96.3	0.0	40.1
POLAR	baseline	Off	Off	On	3.343	2.83	2.66	3.30	4.74	67.5	0.0	39.6
POLAR	baseline	Off	On	Off	3.169	2.70	1.96	3.11	3.13	91.3	92.5	35.6
POLAR	baseline	Off	On	On	3.174	2.71	2.01	3.12	3.13	51.3	30.0	35.6
POLAR	baseline	On	Off	Off	3.332	2.81	5.56	3.28	6.54	85.0	0.0	33.3
POLAR	baseline	On	Off	On	3.361	2.85	2.89	3.35	4.97	91.3	2.5	34.5
POLAR	baseline	On	On	Off	3.178	2.72	1.98	3.12	3.14	73.8	58.8	31.8
POLAR	baseline	On	On	On	3.182	2.73	1.97	3.12	3.14	56.3	21.3	31.2
POLAR	weak	Off	Off	Off	3.333	2.82	2.86	3.28	5.35	88.8	0.0	39.0
POLAR	weak	Off	Off	On	3.348	2.90	3.76	3.38	5.24	93.8	0.0	37.9
POLAR	weak	Off	On	Off	3.179	2.74	1.95	3.12	3.14	70.0	51.3	36.2
POLAR	weak	Off	On	On	3.187	2.74	1.99	3.13	3.15	41.3	41.3	36.2
POLAR	weak	On	Off	Off	3.335	2.82	3.47	3.28	5.56	91.3	0.0	32.8
POLAR	weak	On	Off	On	3.353	2.82	2.58	3.32	4.81	96.3	1.3	32.8
POLAR	weak	On	On	Off	3.178	2.74	1.95	3.12	3.14	33.8	32.5	30.7
POLAR	weak	On	On	On	3.183	2.74	1.94	3.13	3.15	81.3	78.8	30.7
POLAR	strong	Off	Off	Off	3.338	2.85	3.79	3.28	5.83	98.8	1.3	37.3
POLAR	strong	Off	Off	On	3.360	2.85	2.99	3.32	5.01	91.3	0.0	37.9
POLAR	strong	Off	On	Off	3.172	2.72	1.96	3.11	3.13	80.0	52.5	35.8

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Table 7: Complete Results of the 120-Run Ablation Sweep (continued).

Type	Reg	Dist	Mem	Win	Val	Clean2K	Clean64K	Junk2K	Junk64K	Ndl2K	Ndl64K	MFU
POLAR	strong	Off	On	On	3.181	2.72	1.94	3.12	3.15	76.3	68.8	34.6
POLAR	strong	On	Off	Off	3.337	2.87	3.22	3.28	5.52	83.8	1.3	32.8
POLAR	strong	On	Off	On	3.347	2.82	2.64	3.30	4.88	97.5	16.3	32.2
POLAR	strong	On	On	Off	3.179	2.73	1.94	3.12	3.14	86.3	83.8	30.9
POLAR	strong	On	On	On	3.182	2.72	1.94	3.12	3.14	25.0	23.8	30.9
POLAR	discrete	Off	Off	Off	3.338	2.82	2.80	3.29	5.12	93.8	2.5	37.9
POLAR	discrete	Off	Off	On	3.349	2.87	2.90	3.33	5.25	93.8	6.3	37.9
POLAR	discrete	Off	On	Off	3.184	2.75	2.03	3.12	3.14	76.3	73.8	36.4
POLAR	discrete	Off	On	On	3.178	2.71	1.92	3.12	3.15	31.3	18.8	35.8
POLAR	discrete	On	Off	Off	3.333	2.80	4.13	3.28	6.12	97.5	0.0	32.6
POLAR	discrete	On	Off	On	3.351	2.87	2.72	3.33	4.89	85.0	2.5	33.3
POLAR	discrete	On	On	Off	3.189	2.74	1.96	3.13	3.15	87.5	53.8	31.8
POLAR	discrete	On	On	On	3.192	2.74	2.07	3.13	3.15	21.3	26.3	31.7
POLAR	zipfian	Off	Off	Off	3.338	2.85	4.83	3.28	6.55	96.3	11.3	37.9
POLAR	zipfian	Off	Off	On	3.350	2.83	2.58	3.32	4.63	88.8	3.8	39.4
POLAR	zipfian	Off	On	Off	3.171	2.70	1.92	3.11	3.13	52.5	63.8	36.2
POLAR	zipfian	Off	On	On	3.180	2.75	1.95	3.12	3.15	46.3	40.0	35.2
POLAR	zipfian	On	Off	Off	3.340	2.83	3.77	3.28	5.64	98.8	1.3	33.8
POLAR	zipfian	On	Off	On	3.352	2.85	3.13	3.36	5.23	90.0	0.0	33.5
POLAR	zipfian	On	On	Off	3.177	2.72	1.94	3.12	3.14	91.3	67.5	31.8
POLAR	zipfian	On	On	On	3.192	2.74	2.00	3.13	3.15	40.0	26.3	31.8
NOPE	baseline	Off	Off	Off	3.224	2.76	3.71	3.16	6.68	97.5	1.3	41.5
NOPE	baseline	Off	Off	On	3.219	2.75	2.76	3.17	5.49	92.5	10.0	36.5
NOPE	baseline	Off	On	Off	3.140	2.66	2.34	3.09	3.23	97.5	16.3	36.8
NOPE	baseline	Off	On	On	3.150	2.70	2.14	3.09	3.24	81.3	0.0	34.2
NOPE	baseline	On	Off	Off	3.209	2.74	2.99	3.15	6.13	90.0	18.8	32.5
NOPE	baseline	On	Off	On	3.212	2.77	2.90	3.17	5.65	81.3	0.0	30.3
NOPE	baseline	On	On	Off	3.149	2.67	2.35	3.10	3.28	80.0	16.3	30.0
NOPE	baseline	On	On	On	3.147	2.68	2.09	3.09	3.23	88.8	21.3	28.2
NOPE	weak	Off	Off	Off	3.214	2.77	3.10	3.16	6.34	95.0	0.0	40.9

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Table 7: Complete Results of the 120-Run Ablation Sweep (continued).

Type	Reg	Dist	Mem	Win	Val	Clean2K	Clean64K	Junk2K	Junk64K	Ndl2K	Ndl64K	MFU
NOPE	weak	Off	Off	On	3.222	2.75	3.05	3.19	5.81	81.3	0.0	36.1
NOPE	weak	Off	On	Off	3.147	2.68	2.07	3.09	3.19	88.8	21.3	37.5
NOPE	weak	Off	On	On	3.142	2.68	2.12	3.08	3.23	82.5	0.0	33.7
NOPE	weak	On	Off	Off	3.215	2.75	4.87	3.16	12.48	95.0	5.0	32.0
NOPE	weak	On	Off	On	3.221	2.73	2.63	3.18	4.84	81.3	0.0	29.3
NOPE	weak	On	On	Off	3.147	2.67	2.48	3.09	3.23	97.5	1.3	30.8
NOPE	weak	On	On	On	3.148	2.68	2.14	3.09	3.24	93.8	1.3	28.4
NOPE	strong	Off	Off	Off	3.231	2.74	4.88	3.18	11.85	90.0	2.5	38.4
NOPE	strong	Off	Off	On	3.214	2.72	2.68	3.17	5.28	82.5	0.0	36.4
NOPE	strong	Off	On	Off	3.144	2.68	2.37	3.09	3.24	96.3	6.3	35.7
NOPE	strong	Off	On	On	3.150	2.70	2.10	3.10	3.25	93.8	7.5	33.8
NOPE	strong	On	Off	Off	3.207	2.72	3.13	3.15	6.20	73.8	0.0	31.4
NOPE	strong	On	Off	On	3.214	2.74	2.49	3.17	4.36	86.3	8.8	28.9
NOPE	strong	On	On	Off	3.146	2.67	2.24	3.09	3.23	83.8	17.5	29.1
NOPE	strong	On	On	On	3.147	2.68	2.16	3.09	3.28	96.3	0.0	27.5
NOPE	discrete	Off	Off	Off	3.202	2.75	3.33	3.15	6.23	85.0	1.3	38.8
NOPE	discrete	Off	Off	On	3.234	2.79	3.11	3.19	5.47	70.0	18.8	37.0
NOPE	discrete	Off	On	Off	3.145	2.68	2.14	3.09	3.27	98.8	16.3	36.0
NOPE	discrete	Off	On	On	3.145	2.68	2.10	3.09	3.25	95.0	3.8	34.3
NOPE	discrete	On	Off	Off	3.212	2.88	3.04	3.16	6.88	42.5	0.0	31.2
NOPE	discrete	On	Off	On	3.211	2.78	2.73	3.16	5.28	82.5	1.3	29.1
NOPE	discrete	On	On	Off	3.145	2.68	2.27	3.08	3.26	91.3	11.3	30.3
NOPE	discrete	On	On	On	3.149	2.68	2.17	3.08	3.32	83.8	0.0	28.6
NOPE	zipfian	Off	Off	Off	3.209	2.71	3.18	3.15	6.40	90.0	0.0	40.8
NOPE	zipfian	Off	Off	On	3.219	2.75	2.71	3.17	5.44	77.5	1.3	36.1
NOPE	zipfian	Off	On	Off	3.141	2.67	2.55	3.08	3.38	96.3	0.0	38.0
NOPE	zipfian	Off	On	On	3.150	2.69	2.11	3.09	3.25	90.0	3.8	33.7
NOPE	zipfian	On	Off	Off	3.207	2.77	3.32	3.15	7.09	82.5	0.0	31.3
NOPE	zipfian	On	Off	On	3.227	2.79	2.91	3.18	5.94	93.8	11.3	30.2
NOPE	zipfian	On	On	Off	3.146	2.68	2.39	3.10	3.25	96.3	21.3	29.6

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Table 7: Complete Results of the 120-Run Ablation Sweep (continued).

Type	Reg	Dist	Mem	Win	Val	Clean2K	Clean64K	Junk2K	Junk64K	Ndl2K	Ndl64K	MFU
NOPE	zipfian	On	On	On	3.148	2.68	2.17	3.09	3.28	90.0	6.3	28.7
ROPE	baseline	Off	Off	Off	3.159	2.85	3.36	3.09	5.21	42.5	0.0	40.4
ROPE	baseline	Off	Off	On	3.163	2.86	3.79	3.17	5.86	15.0	0.0	37.2
ROPE	baseline	Off	On	Off	3.145	2.81	2.86	3.09	3.57	73.8	0.0	37.1
ROPE	baseline	Off	On	On	3.151	2.80	2.95	3.12	3.72	13.8	0.0	34.4
ROPE	baseline	On	Off	Off	3.163	3.00	3.47	3.10	5.46	38.8	0.0	33.2
ROPE	baseline	On	Off	On	3.167	2.88	4.06	3.16	5.92	0.0	0.0	30.2
ROPE	baseline	On	On	Off	3.153	2.73	2.45	3.09	3.57	75.0	0.0	31.4
ROPE	baseline	On	On	On	3.154	2.83	2.86	3.12	3.72	32.5	0.0	29.4
ROPE	weak	Off	Off	Off	3.158	2.88	3.46	3.10	5.36	46.3	0.0	41.4
ROPE	weak	Off	Off	On	3.163	2.84	3.96	3.18	5.93	12.5	0.0	37.8
ROPE	weak	Off	On	Off	3.149	2.71	2.46	3.09	3.56	37.5	0.0	36.5
ROPE	weak	Off	On	On	3.156	2.85	2.77	3.12	3.60	23.8	0.0	34.9
ROPE	weak	On	Off	Off	3.155	2.83	3.49	3.10	5.27	38.8	0.0	32.3
ROPE	weak	On	Off	On	3.157	3.05	4.12	3.18	5.86	8.8	0.0	30.5
ROPE	weak	On	On	Off	3.149	2.77	2.76	3.09	3.56	57.5	0.0	31.1
ROPE	weak	On	On	On	3.149	2.77	3.13	3.12	3.80	17.5	0.0	28.0
ROPE	strong	Off	Off	Off	3.165	2.78	3.34	3.10	5.27	30.0	0.0	39.0
ROPE	strong	Off	Off	On	3.169	2.90	3.68	3.19	5.80	31.3	0.0	36.3
ROPE	strong	Off	On	Off	3.158	2.85	2.63	3.10	3.50	51.3	0.0	35.9
ROPE	strong	Off	On	On	3.162	2.86	3.08	3.14	3.76	10.0	0.0	33.2
ROPE	strong	On	Off	Off	3.161	2.85	3.58	3.10	5.28	47.5	0.0	31.1
ROPE	strong	On	Off	On	3.165	2.85	4.03	3.19	5.90	2.5	0.0	29.2
ROPE	strong	On	On	Off	3.155	2.80	2.70	3.10	3.62	61.3	0.0	29.3
ROPE	strong	On	On	On	3.162	2.85	3.08	3.13	3.74	2.5	0.0	28.4
ROPE	discrete	Off	Off	Off	3.165	2.91	3.43	3.11	5.42	22.5	0.0	39.7
ROPE	discrete	Off	Off	On	3.158	2.84	4.05	3.15	5.90	3.8	0.0	36.9
ROPE	discrete	Off	On	Off	3.159	2.83	2.67	3.11	3.56	47.5	0.0	36.3
ROPE	discrete	Off	On	On	3.161	2.79	2.71	3.13	3.74	41.3	0.0	33.7
ROPE	discrete	On	Off	Off	3.171	2.79	3.45	3.12	5.36	51.3	0.0	31.5

Continued on next page

Table 7: Complete Results of the 120-Run Ablation Sweep (continued).

Type	Reg	Dist	Mem	Win	Val	Clean2K	Clean64K	Junk2K	Junk64K	Ndl2K	Ndl64K	MFU
ROPE	discrete	On	Off	On	3.164	2.80	3.74	3.16	5.86	21.3	0.0	30.0
ROPE	discrete	On	On	Off	3.162	2.74	2.47	3.11	3.53	50.0	0.0	30.9
ROPE	discrete	On	On	On	3.154	2.79	2.68	3.11	3.67	18.8	0.0	28.9
ROPE	zipfian	Off	Off	Off	3.163	2.82	3.25	3.10	5.22	62.5	0.0	39.8
ROPE	zipfian	Off	Off	On	3.161	2.94	3.94	3.18	5.81	16.3	0.0	37.1
ROPE	zipfian	Off	On	Off	3.149	2.78	2.70	3.09	3.63	46.3	0.0	36.7
ROPE	zipfian	Off	On	On	3.151	2.80	2.63	3.11	3.68	2.5	0.0	33.9
ROPE	zipfian	On	Off	Off	3.161	2.78	3.38	3.10	5.32	47.5	0.0	32.6
ROPE	zipfian	On	Off	On	3.160	3.01	4.11	3.17	5.81	17.5	0.0	30.2
ROPE	zipfian	On	On	Off	3.149	2.73	2.39	3.09	3.55	62.5	0.0	29.8
ROPE	zipfian	On	On	On	3.152	2.79	2.99	3.12	3.78	26.3	0.0	28.5

F Retrieval Breakdown

NIAH Retrieval Depth-Sweep Heatmaps Across Lengths (2K -> 256K) and Depths (10%, 50%, 90%)

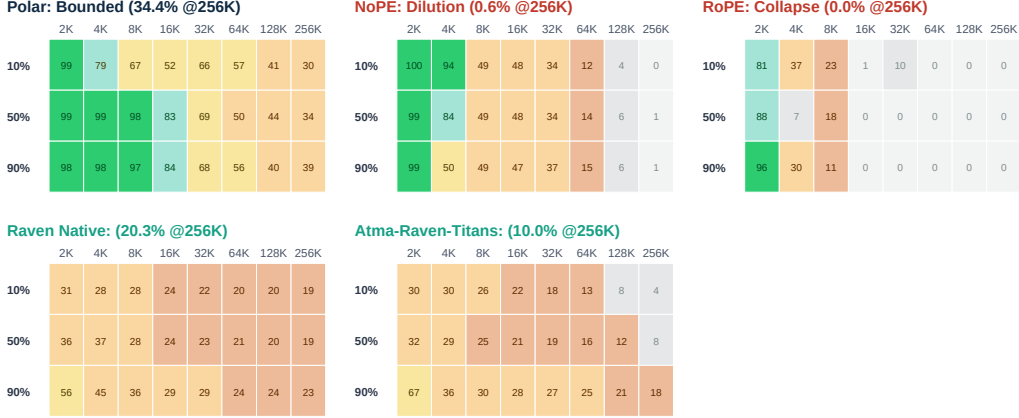


Figure 6: Retrieval depth sweeps for the five promoted models. Each cell is teacher-forced token accuracy.

Model	Haystack	2K	4K	8K	16K	32K	64K	128K	256K
Polar	Synthetic	99.6	98.4	98.5	97.0	94.9	92.3	80.3	68.4
	FinePDFs	98.6	86.3	77.9	49.1	40.7	16.3	3.2	0.3
NoPE	Synthetic	99.5	98.7	98.9	94.9	70.3	26.5	10.2	1.2
	FinePDFs	99.3	52.9	0.0	0.0	0.0	0.0	0.0	0.0
RoPE	Synthetic	81.7	44.9	33.1	0.8	6.8	0.1	0.0	0.0
	FinePDFs	94.8	4.7	0.2	0.0	0.0	0.0	0.0	0.0
Atma-Raven-Titans	Synthetic	64.8	60.7	51.1	46.9	41.7	35.8	25.9	19.3
	FinePDFs	23.2	2.6	0.7	0.2	1.4	0.5	1.6	0.7
Raven Native	Synthetic	55.9	57.7	51.4	48.3	44.3	40.9	38.9	35.1
	FinePDFs	22.1	15.4	8.2	3.3	5.3	2.9	3.7	5.4

Table 8: Zero-shot retrieval accuracy (%) by haystack type and context length, averaged over needle depths.

Figure 6 exposes depth-wise variation, while Table 8 aggregates the same retrieval evaluation by haystack and context length.

G Short-Context Quality and Systems

Group	Model	LAMBADA	HellaSwag	PIQA	WinoGrande	ARC-E	ARC-C	OBQA	BoolQ	Mean
Matched	NoPE	30.16	37.88	66.27	51.93	50.53	32.44	31.60	60.37	45.15
Matched	RoPE	30.25	37.19	66.76	50.67	49.12	29.43	32.40	60.95	44.60
Matched	Polar	28.47	36.19	66.87	51.62	49.12	26.09	31.80	56.21	43.29
Raven/AdamW	Atma-Raven-Titans	26.00	34.12	65.56	52.49	48.07	27.76	29.00	61.41	43.05
Raven/AdamW	Raven Native	27.48	33.56	64.69	51.70	49.12	27.09	28.80	61.93	43.05

Table 9: Task-level zero-shot accuracy (%) on the eight short-context controls (22,939 examples per model). LAMBADA, WinoGrande, and BoolQ use standard accuracy; HellaSwag, PIQA, ARC-Easy, ARC-Challenge, and OpenBookQA use length-normalized accuracy. Mean is the unweighted average of the eight displayed metrics; bold is best within each comparison group.

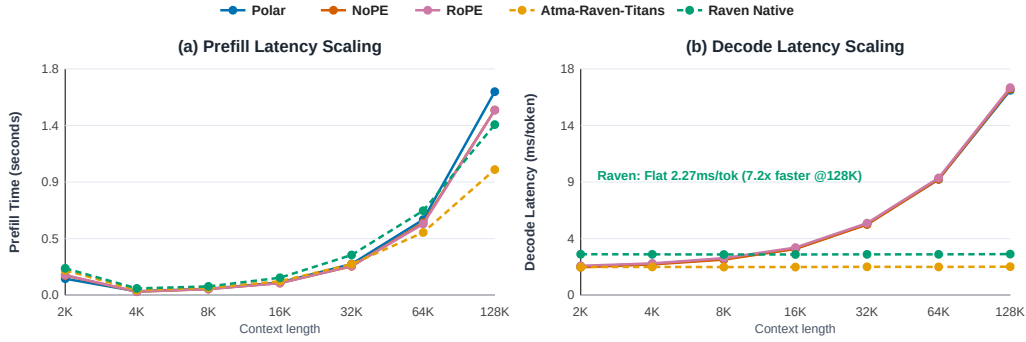


Figure 7: Single-sequence serving on one L40S. Attention decode scales with KV-cache length; recurrent Raven uses fixed state.

Group	Model	Base mean (%)	128K prefill (s)	Decode (ms/token)	Peak alloc. (GiB)	Train MFU (% , 2K)
Matched	NoPE	45.1	1.47	16.4	39.40	42.77
Matched	Polar	43.3	1.62	16.3	39.41	40.44
Matched	RoPE	44.6	1.47	16.5	39.41	42.45
Raven/AdamW	Atma-Raven-Titans	43.1	1.00	2.27	8.62	30.92
Raven/AdamW	Raven Native	43.0	1.36	3.27	6.46	21.53

Table 10: Short-context quality and serving trade-offs. Serving uses one sequence and one timing sample; it is descriptive rather than an uncertainty estimate.

Table 9 gives the exact per-task controls shown in main-text Figure 3. Figure 7 shows the serving-length trend, and Table 10 summarizes the quality-systems trade-off.

H Systems Implementation and Optimization Breakdown

The inference stack uses separate kernels for full-context prefill, paged autoregressive decode, and recurrent-state updates. Table 11 distinguishes what each optimization changes. In particular, streaming removes quadratic score storage but not quadratic prefill arithmetic, while the recurrent Raven path changes the state representation itself and therefore has different scaling.

Path	Optimization	Operational effect
Polar prefill	Block-streamed K, V tiles with online (M, L, Q_2, S) accumulation	Avoids materializing $T \times T$ scores; retains full-context $O(T^2)$ arithmetic.
Polar decode	Paged KV-cache kernel grouping the four query heads that share each KV head	Reuses KV loads across the GQA group; decode remains $O(T)$ in cache length.
Gated-delta step	Fused in-place decay, prediction, delta write, and readout	Avoids materializing per-step intermediate states and preserves $O(1)$ recurrent state size.
Ragged prefill	Packed prompts with grouped compatible tile shapes	Reduces padding work on heterogeneous prompt batches.
Compiler boundary	Recurrent scan and custom kernels exposed as opaque operations	Prevents compiler graph breaks around the stateful update while retaining explicit numerical tests.

Table 11: Implementation-level optimization breakdown. Complexity statements describe the sequence-mixing path and exclude shared MLP and embedding costs.

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Single-sequence scaling. Table 12 gives the underlying single-sequence measurements. On one NVIDIA L40S, attention decode increases from approximately 2.3 ms/token at 2K to 16.3–16.5 ms/token at 128K because every step reads a longer KV cache. The recurrent models remain nearly flat because they update fixed-size state. Prefill is less monotonic at short lengths because launch overhead and kernel occupancy dominate before the matrix operations reach their efficient regime. These timings use batch size 1, 32 generated tokens, and one recorded timing sample per cell; they are descriptive measurements, not confidence intervals.

Table 12: Detailed prefill and decode scaling from 2K to 128K tokens on one NVIDIA L40S GPU (batch size 1, 32 generated tokens).

Model	Metric	2K	4K	8K	16K	32K	64K	128K
Polar	Prefill latency (s)	0.131	0.031	0.051	0.101	0.248	0.601	1.619
	Prefill throughput (tok/s)	15,640	133,884	160,145	161,551	132,091	109,050	80,959
	Decode latency (ms/tok)	2.27	2.49	2.88	3.70	5.66	9.22	16.30
	Decode throughput (tok/s)	441.2	402.3	346.6	270.1	176.8	108.5	61.3
NoPE	Prefill latency (s)	0.157	0.028	0.049	0.097	0.230	0.576	1.473
	Prefill throughput (tok/s)	13,070	146,568	166,504	168,510	142,319	113,810	89,013
	Decode latency (ms/tok)	2.22	2.45	2.83	3.68	5.62	9.22	16.40
	Decode throughput (tok/s)	450.1	408.8	353.0	271.7	177.8	108.4	61.0
RoPE	Prefill latency (s)	0.160	0.028	0.049	0.096	0.232	0.566	1.472
	Prefill throughput (tok/s)	12,761	147,089	168,287	170,936	141,249	115,786	89,015
	Decode latency (ms/tok)	2.34	2.53	2.95	3.79	5.73	9.33	16.51
	Decode throughput (tok/s)	427.4	395.3	338.7	263.7	174.5	107.2	60.6
Atma-Raven-Titans	Prefill latency (s)	0.198	0.041	0.059	0.111	0.247	0.499	0.999
	Prefill throughput (tok/s)	10,334	100,989	137,903	147,434	132,916	131,464	131,195
	Decode latency (ms/tok)	2.24	2.25	2.24	2.24	2.26	2.25	2.27
	Decode throughput (tok/s)	446.6	445.1	446.0	447.1	442.6	444.8	440.3
Raven Native	Prefill latency (s)	0.214	0.054	0.070	0.139	0.320	0.672	1.357
	Prefill throughput (tok/s)	9,558	75,738	116,378	117,605	102,354	97,527	96,559
	Decode latency (ms/tok)	3.26	3.25	3.24	3.24	3.25	3.25	3.27
	Decode throughput (tok/s)	306.4	307.5	308.2	308.3	307.6	307.5	306.1

Production-framework stress comparison. To test whether the custom kernels introduce an obvious systems bottleneck at a larger shape, we configured ATMA’s engine with a 9.2B-parameter stress shape and compared it with vLLM 0.25.0 serving Qwen3.5-9B on the same L40S in BF16 (Kwon et al., 2023). The models are not weight- or architecture-matched, so Table 13 is a backend stress comparison rather than a model-quality or universal serving claim. ATMA trails on short homogeneous prefill, is similar on long heterogeneous prefill, and leads the recorded high-batch decode cells.

Phase	Batch/workload	ATMA engine	vLLM/Qwen3.5-9B	Ratio
Prefill	Homogeneous $B = 8, T = 512$ (4,096 tok)	11,947 tok/s	12,772 tok/s	0.94×
	Heterogeneous short-heavy (816 tok)	5,752 tok/s	8,002 tok/s	0.72×
	Heterogeneous mixed (4,384 tok)	12,779 tok/s	11,996 tok/s	1.07×
	Heterogeneous long-tail (8,192 tok)	12,251 tok/s	11,721 tok/s	1.05×
Decode, context 512	Batch 128	2,939 tok/s	2,247 tok/s	1.31×
	Batch 256	4,501 tok/s	2,921 tok/s	1.54×
	Batch 512	5,919 tok/s	2,558 tok/s	2.31×
	Heterogeneous $B = 128$ (context 64–1536)	2,950 tok/s	2,403 tok/s	1.23×

Table 13: Recorded systems-scale throughput for ATMA’s 9.2B stress shape and vLLM 0.25.0 serving Qwen3.5-9B on one L40S (BF16, greedy sampling). Ratio is ATMA/vLLM.

I Stress and Environment Diagnostics

FEA Structural Stress Probing: Local Block Secant Gains G Across Length (2K -> 256K)

Polar (ATMA): Flat Gains Across Length									RoPE: Representational Attenuation									NoPE (L40S mbs16): Block 7 Yield Locus								
	2K	4K	8K	16K	32K	64K	128K	256K		2K	4K	8K	16K	32K	64K	128K	256K		2K	4K	8K	16K	32K	64K	128K	256K
B0	0.86	0.86	0.86	0.86	0.86	0.86	0.86	0.86	B0	0.94	0.94	0.94	0.94	0.94	0.94	0.94	0.94	B0	0.91	0.91	0.91	0.91	0.91	0.91	0.91	0.91
B1	0.78	0.78	0.78	0.78	0.78	0.78	0.78	0.78	B1	0.92	0.92	0.92	0.92	0.92	0.92	0.92	0.92	B1	0.88	0.88	0.88	0.88	0.88	0.88	0.88	0.88
B2	0.96	0.95	0.95	0.95	0.96	0.96	0.96	0.96	B2	1.01	1.00	0.99	0.98	0.98	0.98	0.98	0.98	B2	1.02	1.02	1.02	1.02	1.02	1.03	1.03	1.03
B3	0.90	0.89	0.89	0.89	0.88	0.88	0.88	0.87	B3	1.02	1.02	1.02	1.02	1.02	1.02	1.02	1.02	B3	0.98	0.98	0.98	0.97	0.97	0.97	0.97	0.97
B4	0.87	0.87	0.86	0.86	0.86	0.85	0.85	0.85	B4	1.05	1.07	1.08	1.07	1.06	1.04	1.03	1.03	B4	0.96	0.96	0.96	0.96	0.96	0.96	0.96	0.95
B5	0.86	0.85	0.85	0.84	0.84	0.84	0.84	0.84	B5	0.94	0.93	0.93	0.93	0.93	0.93	0.93	0.92	B5	0.94	0.94	0.94	0.94	0.95	0.96	0.96	0.96
B6	1.02	1.02	1.10	1.05	1.07	1.09	1.10	1.11	B6	1.09	1.06	1.04	1.02	1.01	1.01	1.00	1.00	B6	1.13	1.11	1.09	1.09	1.08	1.09	1.09	1.10
B7	0.98	0.97	0.97	0.97	0.97	0.97	0.97	0.97	B7	1.01	1.01	1.00	1.00	1.00	1.00	0.99	0.99	B7	1.02	1.02	1.02	1.03	1.09	1.20	1.31	1.38
B8	0.96	0.96	0.95	0.95	0.95	0.95	0.95	0.94	B8	1.01	1.00	1.00	1.00	0.99	0.99	0.99	0.99	B8	1.01	1.01	1.01	1.00	1.00	1.00	1.00	0.99
B9	0.93	0.92	0.92	0.92	0.92	0.92	0.92	0.91	B9	0.98	0.98	0.98	0.97	0.97	0.97	0.97	0.96	B9	0.99	0.99	0.99	0.98	0.98	0.98	0.98	0.99
B10	1.10	1.10	1.10	1.10	1.10	1.10	1.10	1.09	B10	1.09	1.08	1.07	1.06	1.06	1.05	1.04	1.04	B10	1.16	1.14	1.12	1.12	1.11	1.11	1.11	1.11
B11	1.02	1.02	1.02	1.02	1.02	1.01	1.01	1.01	B11	1.02	1.02	1.02	1.01	1.01	1.01	1.00	1.00	B11	1.04	1.03	1.03	1.02	1.01	1.01	1.00	1.00
B12	1.02	1.01	1.01	1.01	1.01	1.00	1.00	1.00	B12	1.02	1.02	1.01	1.01	1.01	1.00	1.00	1.00	B12	1.04	1.04	1.03	1.02	1.01	1.01	1.01	1.00
B13	1.01	1.01	1.00	1.00	1.00	1.00	1.00	1.00	B13	1.02	1.01	1.01	1.00	1.00	1.00	1.00	1.00	B13	1.04	1.04	1.06	1.08	1.09	1.07	1.06	1.06
B14	1.43	1.44	1.45	1.44	1.41	1.39	1.38	1.37	B14	1.32	1.31	1.29	1.28	1.27	1.26	1.25	1.25	B14	1.39	1.37	1.35	1.35	1.30	1.26	1.24	1.23
B15	1.23	1.23	1.22	1.21	1.20	1.19	1.18	1.17	B15	1.21	1.20	1.19	1.19	1.18	1.17	1.17	1.16	B15	1.18	1.18	1.17	1.14	1.12	1.11	1.09	1.08

Figure 8: Block-wise sampled secant gains over 64 nested FinePDFs documents and 64 perturbation directions per document. The probe localizes checkpoint-specific changes but does not estimate a global Lipschitz constant.

Figure 8 reports the sampled gains. For each block f_i , the diagnostic samples

$$G_i = \frac{\|f_i(x + \delta x) - f_i(x)\|_{\text{rms}}}{\|\delta x\|_{\text{rms}}}, \quad \|\delta x\|_{\text{rms}} = 0.02\|x\|_{\text{rms}}. \quad (10)$$

The maximum sampled gain is a lower bound on the local operator norm along sampled directions; it is not a tight bound on the block Lipschitz constant. In the observed NoPE collapse, changes first concentrate near blocks 6–7 and later propagate through the residual stream. Polar’s sampled gains and activation moments remain comparatively flat through 256K.

I.1 Environment-Sensitivity Diagnostic Journey

The sensitivity finding emerged sequentially. Stage I used L4 GPUs, followed by the first Stage II 10B NoPE control on L4. The anomaly became visible only after the remaining Stage II training moved to L40S: short-context convergence remained nearly unchanged while the 256K endpoint diverged. Table 14 records the subsequent elimination sequence.

Step	Observation or intervention	Diagnostic implication
L4 baseline	Stage I and the first Stage II 10B NoPE control used L4, microbatch 4; 256K NLL was 4.07.	Established the original extrapolation behavior, but not a replicated device-class estimate.
L40S transition	The promoted NoPE run used L40S, microbatch 16; 2K validation ended at 2.8126 versus 2.8160 on L4, but 256K NLL rose to 10.77.	Revealed that nearly coincident short-context curves did not predict the long-context endpoint.
Evaluation replay	L4 and L40S checkpoints were re-evaluated under PyTorch 2.12 and 2.13 with unchanged scores.	Excluded evaluation-version mismatch for these checkpoints.
Microbatch control	NoPE was retrained on L40S with microbatch 4; 2K validation ended at 2.8128 and 256K NLL at 9.90.	Excluded microbatch size alone as the explanation in this comparison.
Deterministic fixtures	Synthetic 2K stream fixtures across the tested versions and microbatches showed no gradient or loss divergence.	Found no short-context execution discrepancy; did not test the full stochastic training trajectory.

Table 14: Diagnostic journey from the L4 control to the L40S follow-ups. Each implication is limited to the intervention actually performed.

All five headline Stage II models were trained in the same L40S software and hardware environment, so their primary comparison is internally matched. The cross-device observation nevertheless motivates a broader audit principle: long-context extrapolation claims should report device class, kernels, seeds, and extended-context diagnostics rather than infer robustness from short-context loss alone. The present evidence does not identify CUDA reduction order, floating-point accumulation, or any other single mechanism as causal, because the L4/L40S intervention was not randomized or replicated across multiple seeds and kernel configurations.